

CSCE 650 VIRTUAL REALITY · MODULE 07 · LECTURE NOTES

Cybersickness

Why people feel ill, and what actually helps.

MODULE OUTCOMES

- Explain sensory conflict theory and its predictions.
- Explainvection and why it is the central mechanism.
- Identify the specific design choices that provoke sickness.
- Explain individual variation and why your tolerance is misleading.
- Apply and evaluate the standard mitigations.

1 The mechanism

SENSORY CONFLICT THEORY

The eyes report that you are moving. The vestibular system reports that you are not. The brain receives two signals it has every reason to trust, and they contradict each other.

The leading explanation for why this produces nausea is evolutionary. For most of human history, the commonest cause of a sustained mismatch between visual and vestibular signals was a neurotoxin disrupting the nervous system — and the adaptive response to suspected poisoning is to empty the stomach.

It accounts for the observations well. Passengers are sicker than drivers, because the driver anticipates the motion and has control. Closing the eyes helps, because it removes one of the conflicting signals. Acclimatisation occurs, because the brain learns the conflict is not poison.

It is a theory rather than a settled mechanism. Postural instability theory — that sickness arises from prolonged failure to maintain postural control — is a serious competitor with its own supporting evidence.

Fortunately the design implications are substantially the same either way, so the dispute does not need resolving in order to build comfortable experiences.

VECTION IS THE SPECIFIC CULPRIT

Vection is the illusion of self-motion produced by visual motion alone. The familiar example is sitting in a stationary train and feeling yourself move when the train on the adjacent platform pulls away.

It is strongest in the peripheral visual field, which Module 02 established is specialised for motion detection rather than detail. Large, coherent motion across the periphery is the most potent stimulus.

In VR, any camera motion the user did not physically initiate produces vection — and the vestibular system flatly contradicts it, because the user is standing still.

So the core problem is visually induced motion without corresponding physical motion, and essentially every mitigation in §3 works by reducing either the strength of the vection or the size of the conflict.

Provocation	Why it is provocative
Camera motion the user did not cause	Maximal conflict. The visual system reports motion and the vestibular system reports nothing whatsoever. Cutscenes that move the camera are the worst case.
Acceleration rather than constant velocity	The vestibular system measures acceleration specifically — the semicircular canals measure angular acceleration and the otoliths linear acceleration. At constant velocity it reports nothing, so there is far less to contradict. This is the single most actionable fact in the module and it underlies most of Module 08.
Rotation that is not the user's head rotation	The semicircular canals are extremely sensitive to rotation, and visually rotating the world while the head is still is among the most reliably sickening things possible.
Latency and dropped frames	Module 06. These create conflict <i>even when the user is moving their own head</i> , which is otherwise the safe case.
Tilting or moving the horizon	The otoliths sense gravity, so the horizon's orientation is directly contradicted.
Large coherent motion in the periphery	Maximal vection, per the callout above.

2 Individual variation

SUSCEPTIBILITY VARIES ENORMOUSLY, AND YOU ARE AN OUTLIER

Some people experience no discomfort from content that makes others ill within two minutes. The range across individuals is very wide — wider than for almost any other perceptual response you will design around.

Known correlates include prior VR experience, migraine history, age, inner-ear conditions, and — reported consistently across studies, with the mechanism still debated — sex.

Acclimatisation is the largest single factor and the most relevant one. People who use VR regularly develop substantial tolerance over weeks, and in doing so they lose the ability to judge what a first-time user experiences. The adaptation is real, durable, and invisible from the inside.

Which is exactly why VR developers ship uncomfortable VR. You spend every working day in a headset. **You are, almost by definition, the most acclimatised person who will ever use your application,** and your judgement of its comfort is therefore the least reliable available. This is not a failure of care; it is a structural problem with the role.

- **Test on novices.** Someone who has never worn a headset is the only person who can tell you what a first experience is like, and that experience determines whether most people continue.
- **Use a standard instrument** — the Simulator Sickness Questionnaire, or the shorter VR Sickness Questionnaire. Both are free and validated, and using one makes your results comparable to the published literature rather than being an impression.
- **Administer it before as well as after.** Some participants arrive with a headache, and without a baseline you will attribute it to your application.
- **Report every participant, including the one who stopped after ninety seconds. Averaging away a dropout is the central dishonesty in comfort reporting** — the person who could not continue is the most informative participant you had, and excluding them produces a mean that describes only the people who tolerated it.
- **Have an explicit stopping rule and state it in advance.** Participants stop whenever they wish, without explanation, and nobody is encouraged to continue through discomfort. This is basic research ethics and it also produces better data.

3 Mitigation

Mitigation	Strength of evidence	Notes
Hit the frame rate; minimise latency	Strong	The precondition for everything else. No comfort technique compensates for a dropped frame.
Avoid camera motion the user did not initiate	Strong	Including cutscenes, forced perspective changes, and snapping on tracking loss (Module 05).
Avoid acceleration; prefer constant velocity	Strong	If you must accelerate, do it quickly — a short sharp change is less provocative than a long gentle one, which is counterintuitive and well supported.
Teleport rather than smooth locomotion	Strong	Module 08. The largest single design decision available.
Vignette during movement	Moderate	Darkening the periphery reduces vection at its strongest point. Widely used and genuinely effective.
A stable reference frame	Moderate, well supported	See below.
Snap turning rather than smooth turning	Moderate to strong	Removes the visually induced rotation that the canals object to most.
Reduce field of view during motion	Moderate	The same mechanism as the vignette.
A fixed rest frame or grid overlay	Mixed	Studied repeatedly with inconsistent results. Worth trying; do not assume.

A STABLE REFERENCE FRAME IS CHEAP AND UNDERUSED

Give the user something that moves with them and remains stationary in their view — a cockpit, a vehicle interior, a helmet rim, the edge of a visor.

It supplies a visual reference consistent with the vestibular signal — part of the visual field is saying 'you are not moving relative to this', which is what the inner ear is also saying. The conflict is reduced rather than merely masked.

This is a large part of why cockpit-based games are so much more comfortable than free first-person movement, and the effect is substantial rather than marginal. Driving and flying games were comfortable in VR years before walking games were.

A virtual nose works. It sounds absurd, users do not consciously notice it when asked, and it measurably reduces sickness — presumably because your real nose is always in the visual field and the brain treats its presence as evidence about self-motion. It is a few triangles and it is worth trying.

DESIGN AROUND IT RATHER THAN MITIGATING AFTERWARDS

Vignettes, comfort modes, and snap turning are patches. They help genuinely, and every one of them is applied to a design that has already created the problem they address.

The comfortable experiences are comfortable by construction. The user is seated; or stationary and the world comes to them; or in a cockpit moving at constant velocity; or moving by teleport. No patch was needed because no conflict was created.

Decide the locomotion model first, before art, before mechanics, before anything. It determines what the application can be, and changing it later is a redesign rather than an adjustment (Module 08).

And offer options. Given the variation in §2, no single choice suits everyone — the comfort settings menu is not a courtesy but **an accessibility feature**, and an application with only one locomotion scheme has excluded part of its audience by construction.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
LaValle — Virtual Reality, Chapter 10 (free)	VR sickness: the competing theories, the evidence, and the design implications. The best free treatment.
Kennedy et al. — Simulator Sickness Questionnaire (free, widely reproduced)	The standard instrument of §2. Use it rather than inventing your own scale.
Meta — VR comfort and locomotion guidelines (free)	The practical mitigations with the vendor's own testing behind them.
Rebenitsch & Owen — Review on cybersickness in applications and visual displays (free)	A survey of the evidence for §3's mitigations, including which ones replicate and which do not.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Experience vection deliberately: find a VR scene with large peripheral motion and stand still. Describe the sensation.
- 2 Build a scene with smooth forward movement at constant velocity, and another with acceleration and deceleration. Compare on six people.
- 3 Add a vignette during movement and measure its effect with a questionnaire rather than by impression.

- 4 Add a cockpit or helmet frame and compare comfort against the same motion without one.
- 5 **Add a virtual nose** and test whether anyone notices it, and whether comfort scores change.
- 6 Implement snap turning and smooth turning and compare across participants.
- 7 Administer an SSQ before and after a five-minute session on six people, including at least two novices. Report every score.
- 8 Tilt the horizon by 10 degrees for ten seconds and record how quickly participants ask to stop.
- 9 Write a comfort settings menu for your Project 2 application and justify each option from this module.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 State sensory conflict theory and the evolutionary explanation, and name its main competitor.
- 2 What is vection, where is it strongest, and why does that matter?
- 3 Give six provocations in order, and say which fact about them is most actionable.
- 4 Why does acceleration provoke more than constant velocity?
- 5 Why is a VR developer a poor judge of their own application's comfort?
- 6 Give five rules for testing comfort honestly, and say which is most often violated.
- 7 Rank the mitigations by evidence and name the four that matter most.
- 8 Why does a stable reference frame help, and why does a virtual nose work?
- 9 Why should locomotion be decided first, and why are comfort options an accessibility feature?